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MA3521 — $\mathbf{y}' = A\mathbf{y}+\mathbf{f}(t)$ Exponential and Trig (II) Key Name:

1. Solve $\mathbf{y}' = A\mathbf{y} + e^{-2t}\begin{pmatrix}-7\\-7\end{pmatrix}$, $\mathbf{y}(0) = \begin{pmatrix}4\\0\end{pmatrix}$ and $A = \begin{pmatrix}2 & 3\\3 & 2\end{pmatrix}$ with epairs 5, $\begin{pmatrix}1\\1\end{pmatrix}$; -1, $\begin{pmatrix}1\\-1\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$ $\begin{pmatrix}e^{5t} & e^{-t}\\e^{5t} & -e^{-t}\end{pmatrix}\mathbf{c}$

$\mathbf{y}_c = c_1 e^{5t}\begin{pmatrix}1\\1\end{pmatrix} + c_2 e^{-t}\begin{pmatrix}1\\-1\end{pmatrix}$

(b) $\mathbf{y}_p(t) =$ $e^{-2t}\mathbf{a}$ is a suitable UC form.

(c) Row reducing $\left[\,-2I - A \,\middle|\, \begin{matrix}-7\\-7\end{matrix}\,\right]$ gives the UCs in

$\mathbf{y}_p(t) =$ $e^{-2t}\begin{pmatrix}1\\1\end{pmatrix}$

$(-2I - A)\mathbf{a} = \begin{pmatrix}-7\\-7\end{pmatrix}$

$-2I - A = \begin{pmatrix}-4 & -3\\-3 & -4\end{pmatrix}$

$\mathbf{a} = \begin{pmatrix}1\\1\end{pmatrix}$

(d) Row reducing $\left[\begin{array}{cc|c}1 & 1 & 3\\1 & -1 & -1\end{array}\right]$

$\left[\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix} \mid \mathbf{y}(0) - \mathbf{a}\right]$

$\begin{pmatrix}4\\0\end{pmatrix} - \mathbf{a} = \begin{pmatrix}3\\-1\end{pmatrix}$

$\mathbf{c} = \begin{pmatrix}1\\2\end{pmatrix}$

gives constants $\mathbf{c}$ in

$\mathbf{y}(t) =$ $\begin{pmatrix}e^{5t} + 2e^{-t}\\e^{5t} - 2e^{-t}\end{pmatrix} + e^{-2t}\begin{pmatrix}1\\1\end{pmatrix}$

2. Solve $\mathbf{y}' = A\mathbf{y} + \cos(2t)\begin{pmatrix}5\\5\end{pmatrix}$, $\mathbf{y}(0) = \begin{pmatrix}0\\-2\end{pmatrix}$ and $A = \begin{pmatrix}2 & -1\\3 & -2\end{pmatrix}$ with epairs $1, \begin{pmatrix}1\\1\end{pmatrix}$; $-1, \begin{pmatrix}1\\3\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$ $\boxed{\begin{pmatrix}e^t & e^{-t}\\ e^t & 3e^{-t}\end{pmatrix}\mathbf{c}}$

$\mathbf{y}_c = c_1 e^t \begin{pmatrix}1\\1\end{pmatrix} + c_2 e^{-t}\begin{pmatrix}1\\3\end{pmatrix}$

(b) $\mathbf{y}_p(t) =$ $\boxed{\mathbf{a}\cos(2t) + \mathbf{b}\sin(2t)}$ is a suitable UC form.

(c) Row reducing $\boxed{\left[\begin{array}{cc|c} -A & 2I & \begin{matrix}5\\5\end{matrix}\\ -2I & -A & \begin{matrix}0\\0\end{matrix}\end{array}\right]}$ gives the UCs in

$$\begin{pmatrix}-A & 2I\\ -2I & -A\end{pmatrix}\begin{pmatrix}\mathbf{a}\\\mathbf{b}\end{pmatrix} = \begin{pmatrix}5\\5\\0\\0\end{pmatrix}$$

$$= \begin{pmatrix}-2 & 1 & 2 & 0\\ -3 & 2 & 0 & 2\\ -2 & 0 & -2 & 1\\ 0 & -2 & -3 & 2\end{pmatrix}\begin{pmatrix}\mathbf{a}\\\mathbf{b}\end{pmatrix}$$

$\mathbf{y}_p(t) =$ $\boxed{\begin{pmatrix}-1\\-1\end{pmatrix}\cos(2t) + \begin{pmatrix}2\\2\end{pmatrix}\sin(2t)}$

$$\mathbf{a} = \begin{pmatrix}-1\\-1\end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix}2\\2\end{pmatrix}$$

(d) Row reducing $\boxed{\left[\begin{array}{cc|c} 1 & 1 & 1\\ 1 & 3 & -1\end{array}\right]}$

$$\left[\begin{pmatrix}1 & 1\\ 1 & 3\end{pmatrix} \mid \mathbf{y}(0) - \mathbf{a}\right]$$

$$\begin{pmatrix}0\\-2\end{pmatrix} - \mathbf{a} = \begin{pmatrix}1\\-1\end{pmatrix}$$

$$\mathbf{c} = \begin{pmatrix}2\\-1\end{pmatrix}$$

gives constants $\mathbf{c}$ in

$\mathbf{y}(t) =$ $\boxed{\begin{pmatrix}2e^t - e^{-t}\\ 2e^t - 3e^{-t}\end{pmatrix} + \mathbf{y}_p(t)}$